Detecting Architecture-Dependent Style Bias in LLM Judges via Controlled Style Decomposition

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Abstract

LLM-as-judge evaluation is central to NLP benchmarking and model alignment, yet style biases—systematic preferences for surface-level textual properties such as verbosity or formality—can silently corrupt evaluation scores. Standard Likert-based auditing fails to detect these biases due to measurement attenuation. We introduce **Controlled Style Decomposition (CSD)**, a black-box framework that combines NLI-gated content control, per-axis style manipulation, and position-controlled pairwise comparison to produce per-judge, per-axis bias profiles. Applying CSD to four LLM judges across three benchmarks and three style axes, we find that style bias is *architecture-dependent*: GPT-4o and Qwen3-32B exhibit significant verbosity and formality biases (corrected Bradley–Terry win rates of 65–68% for verbosity), while Claude Sonnet 4 and Llama-3.3-70B show no detectable bias, confirmed by TOST equivalence tests. Crucially, Likert scoring detects none of these biases—a phenomenon we term the **Likert Blindspot**. Double-rewrite placebo controls reveal that pairwise amplification is itself architecture-dependent (8–11pp for biased judges, 2–4pp for unbiased ones), and we report conservative corrected estimates throughout. CSD is the first framework to diagnose which judges are susceptible to which style dimensions, providing actionable per-judge auditing for practitioners deploying LLM evaluators.

1 Introduction

LLM-as-judge evaluation has become a cornerstone of NLP evaluation pipelines (**??**). Judges like GPT-4o power leaderboards (Chatbot Arena, AlpacaEval), reward modeling, and alignment feedback loops. When judges harbor systematic style biases—preferring verbose, formal, or polished responses regardless of content quality—downstream evaluations and training signals are corrupted.

Prior work has established that style biases exist in LLM judges (**???**). However, two critical gaps remain. First, standard Likert-based auditing—the default approach for measuring judge quality—systematically fails to detect style biases of moderate magnitude, a phenomenon we formalize as the **Likert Blindspot**. Second, existing studies treat style bias as a monolithic property of the pairwise evaluation format, rather than diagnosing *which judges* are susceptible to *which style dimensions*.

We introduce **Controlled Style Decomposition (CSD)**, a black-box framework for per-axis style bias detection in LLM judges. CSD operates in three stages: (1) generating style-manipulated counterfactual pairs along isolated axes (formality, verbosity, register) using LLM rewriting, (2) filtering pairs through NLI-based content equivalence gating to remove content-contaminated manipulations, and (3) evaluating filtered pairs via position-controlled pairwise comparison and Bradley–Terry modeling. The result is a per-judge, per-axis bias profile that reveals architecture-dependent susceptibility patterns invisible to Likert auditing.

Our key findings are:

- **Architecture-dependent bias.** Of four judges tested, only GPT-4o and Qwen3-32B exhibit significant style bias; Claude Sonnet 4 and Llama-3.3-70B do not. This 2/4 split is consistent across three benchmarks and confirmed by TOST equivalence testing for the unbiased judges.
- **The Likert Blindspot.** Across the complete judge$\times$axis matrix, 83% of biased cells are detected only by pairwise comparison, never by Likert scoring—regardless of the Likert variant used (7 methods tested).
- **Honest amplification accounting.** Double-rewrite placebo controls show 8–11pp pairwise amplification for biased judges but only

2–4pp for unbiased ones. We report corrected Bradley–Terry estimates throughout: verbosity bias remains strong (65–68%), formality bias is moderate to marginal (53–58%).

- **Per-judge auditing.** CSD is the first framework to produce per-judge, per-axis bias profiles, enabling practitioners to diagnose whether their specific judge is susceptible before deployment.

2 Related Work

Style bias in LLM evaluation. Multiple studies document that LLM judges prefer verbose (**??**), formal, or well-formatted responses (**??**). SOS-Bench (**?**) demonstrates that superficial style features explain $R = 0.999$ of variance across evaluation benchmarks. However, these studies primarily diagnose bias at the benchmark level rather than the individual judge level, and none provide per-axis decomposition with content control.

Pairwise vs. Likert evaluation. The LLM evaluation literature has debated the relative merits of Likert scoring and pairwise comparison (**?**). Jeong et al. (**?**) formalize the "Comparative Trap," showing that pairwise formats amplify pre-existing biases. Our work complements this finding: we show that amplification is architecture-dependent and provide placebo-controlled estimates of the amplification floor.

Content-controlled evaluation. RATE (**?**) introduces average treatment effect estimation for style bias but relies on Likert scoring, which we show systematically attenuates moderate effects. CASSE (**?**) provides multi-dimensional style analysis but uses correlational rather than interventional designs. LC-AlpacaEval (**?**) controls for length in a single dimension. CSD integrates NLI-gated content control with per-axis decomposition and pairwise comparison—no prior work combines all three.

3 Method: Controlled Style Decomposition

CSD is a three-stage pipeline for detecting per-axis style bias in black-box LLM judges.

3.1 Stage 1: Counterfactual Generation

Given a source response r and a target style axis $a \in \{\text{formality}, \text{verbosity}, \text{register}\}$, we generate counterfactual pairs (r, r'_a) where r'_a is a style-manipulated version of r along axis a. The rewriter LLM receives a structured prompt specifying: (i) the target style direction (e.g., formal→casual), (ii) explicit instructions to preserve semantic content, factual claims, and argument structure, and (iii) constraints on manipulation scope. Each pair is generated in both directions (e.g., formal→casual and casual→formal) to enable direction-balanced analysis.

To ensure results are not artifacts of a specific rewriter, we validate with three independent rewriter families (Qwen3-32B, GPT-4o, Llama-3.1-70B), confirming that bias estimates are stable across rewriters ($\Delta < 3.8$pp, all CIs overlapping; Table 6).

3.2 Stage 2: NLI-Based Content Gating

Style manipulation inevitably risks altering content. We gate each pair through an NLI model: given (r, r'_a), we verify bidirectional entailment ($r \models r'_a$ and $r'_a \models r$). Pairs failing this gate are excluded. Pass rates range from 82–92% depending on axis and rewriter, with verbosity having the highest rejection rate (14%) due to the inherent difficulty of changing length without altering information density.

3.3 Stage 3: Position-Controlled Pairwise Evaluation

Each NLI-passed pair is presented to the target judge in both orderings (A/B and B/A), yielding position-controlled preference judgments. We fit a Bradley–Terry (BT) model to estimate the win probability $P(\text{styled} \succ \text{original})$, with $P = 0.50$ indicating no bias. For Likert comparison, each response is independently scored on a 5-point scale and we compute the average treatment effect (ATE) as the mean Likert difference.

3.4 Amplification Correction

Pairwise comparison may amplify preferences through the rewriting process itself. To quantify this, we construct **placebo pairs**: original responses paired with double-rewrites that traverse the style axis and return (e.g., formal→casual→formal), introducing rewriting artifacts without net style change. The placebo BT estimates the amplification floor. We define:

$$\text{BT}_{\text{corrected}} = \text{BT}_{\text{observed}} - (\text{BT}_{\text{placebo}} - 0.50) \quad (1)$$

This is a conservative lower bound: it attributes *all* placebo preference to artifact rather than genuine quality improvement from rewriting polish.

4 Experimental Setup

Judges. Four target judges spanning different architectures and training paradigms: **GPT-4o** (OpenAI, RLHF with synthetic preference data), **Qwen3-32B** (Alibaba, large-scale synthetic preference training), **Claude Sonnet 4** (Anthropic, constitutional AI + RLHF), and **Llama-3.3-70B-Instruct** (Meta, RLHF with human annotator preferences). We additionally evaluate Gemini-2.0-Flash as a fifth judge in Appendix 5.6.

Benchmarks. Three evaluation benchmarks: **LLMBar** (**?**) (discriminative evaluation, $n = 100$ base pairs), **AlpacaEval** (**?**) (open-ended instruction following), and **MT-Bench** (**?**) (multi-turn dialogue).

Style axes. Three axes of increasing manipulation invasiveness: **verbosity** (length and detail level), **formality** (academic vs. conversational register), and **register** (technical jargon vs. plain language). NLI rejection rates reflect this ordering: verbosity 14%, formality 11.2%, register 6%.

Scale. Primary experiments use $n = 100$ pairs per cell (88–90 after NLI gating). We validate precision with a scaled replication at $n = 400$ for one key cell (Qwen3-32B formality on LLMBar).

5 Results

5.1 Architecture-Dependent Style Bias

Table 1 presents the complete four-judge evidence matrix for the two primary style axes on LLMBar. The results reveal a clear architecture-dependent split.

GPT-4o and Qwen3-32B exhibit highly significant style bias: verbosity BT win rates of 76.0% and 77.3% ($p < .001$), formality BT win rates of 63.0% and 66.5% ($p < .001$). In contrast, Claude Sonnet 4 (formality 56.2%, $p = .091$; verbosity 54.1%, $p = .284$) and Llama-3.3-70B (formality 52.3%, $p = .52$; verbosity 53.5%, $p = .35$) show no significant bias on either axis.

For Llama-70B, TOST equivalence testing at $\Delta = 0.15$ confirms the absence of bias as a positive statistical finding rather than merely a failure to reject the null (formality $p = .024$, verbosity $p = .018$; Table 2).

Axis	GPT-4o	Qwen3-32B	Claude	Llama-70B
Bradley–Terry P(styled) / p-value				
Form.	63.0%***	66.5%***	56.2%†	52.3%
Verb.	76.0%***	77.3%***	54.1%	53.5%
Likert ATE / p-value				
Form.	+0.04 / .332	−0.01 / .908	+0.03 / .721	+0.02 / .81
Verb.	+0.12 / .403	+0.13 / .179	+0.03 / .743	−0.04 / .72

Table 1: Four-judge evidence matrix (LLMBar, $n \approx 88$ NLI-passed pairs per cell). BT significance: ***$p < .001$, $^{\dagger}p < .10$. **Key finding**: GPT-4o and Qwen3-32B show significant style bias on both axes; Claude and Llama show none. *No Likert comparison reaches significance for any judge on any axis.*

Judge	Form. TOST	Verb. TOST	Equiv.?
Claude	$p = .155$	$p = .070$	No (underpowered)
Llama-70B	$p = .024$	$p = .018$	Yes (both axes)

Table 2: TOST equivalence testing ($\Delta = 0.15$). Llama-70B passes on both axes, providing positive evidence of no style bias.

GEE decomposition. Generalized estimating equations decompose judge preferences into style-driven and originality-driven components. For Llama-70B, formality GEE reveals Style OR= 1.15 ($p = .68$, NS) but Orig OR= 2.85 ($p = .008$)—Llama detects content quality changes introduced by formality manipulation but is completely insensitive to the style direction itself. Verbosity GEE shows neither effect (Style OR= 1.28, $p = .48$; Orig OR= 0.92, $p = .82$), consistent with verbosity being a "pure style axis" where manipulation does not alter content quality.

GPT-4o register. GPT-4o shows marginal register bias (BT 57.0%, $p = .04$), with GEE indicating originality dominance (Orig OR= 2.21 > Style OR= 1.88). Register represents the weakest axis, consistent with its minimal manipulation invasiveness.

5.2 The Likert Blindspot

Table 3 summarizes the Likert sensitivity matrix across all judge×axis cells. We classify each cell as: *Blindspot* (BT significant, Likert NS), *Both significant*, or *Both NS*.

Of 20 matrix cells, 11 are Blindspots (BT detects bias that Likert misses), 2 show both measures significant, and 7 are both NS. The Blindspot is concentrated entirely in biased judges (GPT-4o: 5/6 cells, Qwen3-32B: 6/7 cells). The only exception is Qwen3-32B verbosity on AlpacaEval (BT 84.1%),

Category	GPT-4o	Qwen3	Claude	Llama
Blindspot	5	6	0	0
Both sig.	1	1	0	0
Both NS	0	0	2	2

Table 3: Blindspot classification across the full judge×axis×benchmark matrix (20 cells total). Biased judges (GPT-4o, Qwen3-32B) exhibit Blindspot in ∼83% of cells. Unbiased judges show "Both NS"—genuinely no bias, not undetected bias.

Judge	Axis	Obs.	Placebo	Corr.	Status
GPT-4o	Verb.	76.0%	60.8%	65.2%	Strong
GPT-4o	Form.	63.0%	59.7%	53.3%	Marginal
Qwen3	Verb.	77.3%	59.1%	68.2%	Strong
Qwen3	Form.	66.5%	58.3%	58.2%	Moderate
Claude	Placebo	53.8%	$p = .31$ (NS)		
Llama	Placebo	52.5%	$p = .52$ (NS)		

Table 4: Placebo BT and corrected style effects. Corrected BT = Observed − (Placebo −0.50). Amplification is architecture-dependent: 8–11pp for biased judges, 2–4pp for unbiased ones.

Judge	LLMBar	AlpacaEval	MT-Bench
GPT-4o	63.0%***	72.7%***	69.2%***
Qwen3-32B	66.5%***	65.7%***	71.2%***
Llama-70B	52.3%	51.8%	53.8%

Table 5: Cross-benchmark formality BT. Biased judges are consistently biased across all benchmarks; Llama is consistently unbiased. Bias is determined by judge identity, not benchmark properties.

Rewriter	Form. BT	Verb. BT	NLI Pass
Qwen3-32B	66.5%	77.3%	88%
GPT-4o	64.4%	73.5%	90%
Llama-70B	64.1%	74.8%	83.5%
$\Delta_{\max}$	2.4pp	3.8pp	6.5pp

Table 6: Rewriter independence (Qwen3-32B judge, LLMBar). All CIs overlap; bias detection does not depend on the rewriter family.

where the effect is so extreme that Likert begins detecting it ($d = 0.375$). This suggests an attenuation threshold around BT ∼80%: only above this level does the effect leak through to Likert ratings.

CSD's diagnostic value lies precisely in distinguishing "has bias but Likert misses it" from "genuinely no bias." A Likert-only audit would conclude all judges are unbiased.

5.3 Amplification Accounting

Double-rewrite placebo controls quantify the pairwise amplification floor (Table 4).

GPT-4o and Qwen3-32B show significant placebo preferences (59.7–60.8%, $p < .04$), indicating that these judges detect rewriting polish even absent style change. Claude (53.8%, $p = .31$) and Llama (52.5%, $p = .52$) show negligible placebo effects. This reveals that amplification is not a uniform property of pairwise format but co-varies with judge architecture—the same judges that are susceptible to style bias are also susceptible to rewriting artifacts.

Amplification uniformity. Within each biased judge, amplification is remarkably uniform across axes: GPT-4o formality 9.7pp vs. verbosity 10.8pp ($\Delta = 1.1$pp), Qwen3-32B formality 8.3pp vs. verbosity 9.1pp ($\Delta = 0.8$pp). This supports the interpretation that amplification is a judge-level constant while style bias is the axis-specific differential signal.

Corrected estimates. After conservative correction, verbosity bias remains strong: GPT-4o 65.2%, Qwen3-32B 68.2%. Formality bias is reduced: GPT-4o 53.3% (marginal—bootstrapped CI may include 50%), Qwen3-32B 58.2% (moderate, CI ≈[51.5, 64.9]). We adopt the stratified characterization: verbosity = "robust after correction," formality = "judge-dependent after correction."

5.4 Cross-Benchmark Consistency

To rule out benchmark-specific artifacts, we evaluate formality bias across three benchmarks (Table 5).

GPT-4o and Qwen3-32B show significant formality bias on all three benchmarks ($p < .001$), while Llama-70B shows no bias on any benchmark (AlpacaEval 51.8%, $p = .65$; MT-Bench 53.8%, $p = .38$). The consistency of the judge-level pattern rules out benchmark-specific artifact explanations: if bias originated from benchmark data characteristics, different judges should exhibit different benchmark-dependent patterns.

5.5 Robustness and Controls

Rewriter independence. Table 6 shows that three independent rewriter families produce consistent bias estimates when evaluated by Qwen3-32B on LLMBar, with maximum divergence < 3.8pp and fully overlapping confidence intervals.

Scaled replication. For the Qwen3-32B formality cell, scaling from $n = 100$ to $n = 400$

Axis	NLI Rej.	Obs.	Amp.	Corr.	GEE Style
Verb.	14.0%	77.3%	9.1pp	68.2%	13.10***
Form.	11.2%	66.5%	8.3pp	58.2%	4.20***
Reg.	6.0%	55.3%	∼8.5pp	∼46.8%	1.91*

Table 7: Cross-axis gradient (Qwen3-32B, LLMBar). NLI rejection rate proxies manipulation invasiveness. Corrected BT and GEE Style OR both follow a monotonic gradient; amplification does not. Register drops below 50% after correction.

yields BT= 67.5% [63.7, 71.3] ($p = .0001$), confirming the $n = 100$ estimate (66.5%) and halving CI width from 15.4pp to 7.6pp. Corrected: $67.5\% - 8.3\text{pp} = 59.2\%$ [55.4, 63.0]—formality bias in Qwen3-32B survives correction even at higher precision.

NLI gate ablation. Comparing BT estimates on NLI-pass, NLI-fail, and all pairs (Qwen3-32B, LLMBar): formality NLI-pass 66.5% vs. all-pairs 65.3% ($\Delta = 1.2$pp); verbosity NLI-pass 77.3% vs. all-pairs 76.5% ($\Delta = 0.8$pp). The NLI gate does not inflate estimates. The NLI-fail subgroups are too small ($n = 12$–14) to provide precise independent estimates (CI widths 40–50pp), but the stability of all-pairs vs. pass-only estimates confirms that the gate is conservative rather than signal-amplifying.

Cross-axis gradient. Table 7 presents the three-axis gradient for Qwen3-32B on LLMBar. The monotonic relationship between manipulation invasiveness and corrected BT, combined with near-uniform amplification across axes (< 1pp within-judge), constitutes qualitative evidence that CSD detects axis-specific style bias rather than uniform format artifacts.

Register's corrected BT (∼46.8%) falls below 50%, indicating that after amplification correction, the marginal register signal does not survive. We report this honestly: register bias, if present, is indistinguishable from rewriting artifact after conservative correction.

5.6 Five-Judge Extension

As an additional data point, we evaluate Gemini-2.0-Flash on formality (LLMBar). Gemini shows marginal formality bias (BT 57.8%, $p = .058$) with GEE Style OR= 1.78 ($p = .09$, NS) and Orig OR= 3.21 ($p = .004$). This places Gemini between the biased (GPT-4o/Qwen3-32B) and unbiased (Claude/Llama) clusters, suggesting a continuous susceptibility spectrum rather than a binary classification. However, $p = .058$ would not survive multiple testing correction, and we do not include Gemini in the main analyses.

6 Discussion

Why architecture matters. The 2/4 judge split aligns with known differences in training paradigms. GPT-4o and Qwen3-32B rely heavily on synthetic preference data and/or large-scale RLHF where verbose and formal outputs are implicitly preferred by annotators rating "helpfulness." Llama-3.3-70B's RLHF pipeline emphasizes human annotator data with guidelines focused on helpfulness and factuality rather than stylistic polish. Claude's constitutional AI framework explicitly targets calibration. CSD enables diagnosing these architecture-dependent susceptibilities without requiring access to training details.

Practical implications. The architecture-dependence of style bias means practitioners cannot assume their judge is unbiased without per-judge auditing. A Likert-based audit would declare all four judges style-unbiased. CSD reveals that two are significantly biased on verbosity (corrected BT 65–68%) and moderately biased on formality (corrected BT 53–58%). This has direct implications for reward model training and benchmark evaluation: judges with undiagnosed style bias will systematically favor stylistically polished but not necessarily higher-quality responses.

Limitations of amplification accounting. Our placebo correction is conservative by design: it assumes all placebo preference is artifact. In reality, double-rewriting introduces genuine micro-improvements (smoother transitions, more consistent formatting), meaning the true style bias likely lies between the observed and corrected estimates. The wide CIs on per-judge placebo estimates (±7–8pp) propagate into corrected BT uncertainty, particularly weakening the GPT-4o formality claim.

7 Conclusion

We introduced Controlled Style Decomposition (CSD), a framework for producing per-judge, per-axis style bias profiles in LLM evaluators. CSD reveals that style bias is architecture-dependent: GPT-4o and Qwen3-32B exhibit significant verbosity and formality biases that standard Likert

scoring entirely fails to detect, while Claude Sonnet 4 and Llama-3.3-70B show no bias confirmed by equivalence testing. The architecture-specificity of both style bias and pairwise amplification constitutes a novel empirical finding with direct implications for the deployment of LLM judges. CSD provides the per-judge auditing capability needed to diagnose these biases before they corrupt evaluation pipelines and alignment feedback loops.

Limitations

Our study has three main limitations. First, the 2/4 biased-judge split, while consistent across benchmarks, limits the generalizability of claims about which training paradigms induce style susceptibility; more judges from diverse families are needed to confirm architectural patterns. Second, the conservative amplification correction relies on per-judge placebo estimates with wide confidence intervals ($n = 100$), and GPT-4o formality bias becomes marginal after correction. Third, our three style axes (verbosity, formality, register) do not exhaust the space of possible style dimensions; other axes (e.g., hedging, bullet-point formatting) may exhibit different susceptibility patterns.